

COMPUTER HARDWARE LAB MANUAL

List of Practicals

1. Identify Internal Components of a Desktop Computer and Observe Their Functions
2. Study & Comparison of CISC and RISC Architecture
3. Identify Motherboard Components (CPU Socket, RAM Slots, Chipset, Expansion Slots and Connectors)
4. Observe and List Functions of Front Panel and Back Panel Ports of a PC
5. Study SMPS Connectors and Measure Output Voltages Using a Digital Multimeter
6. Install and Remove RAM Modules and Verify Detection in BIOS/UEFI
7. Install HDD/SSD and Connect SATA/NVMe Interface Cables
8. Configure BIOS/UEFI Settings and Observe Hardware Detection Parameters
9. Identify Different Types of Memory Modules and Storage Devices
10. Connect and Test Input Devices Such as Keyboard, Mouse and Scanner
11. Install and Configure a Printer and Perform a Test Print
12. Identify Monitor Ports and Configure Display Settings in the Operating System
13. Perform POST and Interpret Beep Codes and Startup Error Messages
14. Troubleshoot Common Hardware Faults Such as No Display, Boot Failure and RAM Errors
15. Assemble and Disassemble a PC Cabinet Following Safety and ESD Precautions

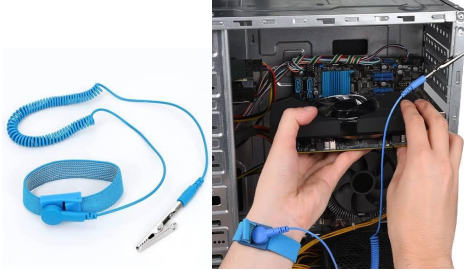
PRACTICAL 1

Identify Internal Components of a Desktop Computer and Observe Their Functions

Aim: To identify the internal components of a desktop computer cabinet and understand the function of each component.

Requirements

- Desktop computer (cabinet opened)
- Screwdriver set
- Anti-static wrist strap



- Torch/inspection light



Theory

A desktop computer cabinet houses several major hardware components that work together to process, store, and transfer data. Identifying each part and understanding its function is the first step towards computer hardware servicing.

Diagram

INTERNAL COMPONENTS OF A DESKTOP COMPUTER

and Their Functions

1 **Power Supply Unit (PSU)**
Converts AC power from the wall outlet to DC power and supplies it to all components.

2 **CPU (Processor)**
The brain of the computer. It executes instructions and performs calculations.

3 **CPU Cooler (Heat Sink & Fan)**
Dissipates heat generated by the CPU to keep it running at safe temperatures.

4 **RAM (Memory)**
Temporarily stores data and instructions that the CPU is currently using for quick access.

5 **Motherboard**
The main circuit board that connects all components and allows them to communicate.

6 **Optical Drive (ODD)**
Reads data from CDs, DVDs or Blu-ray discs and (in some cases) writes data.

7 **Hard Disk Drive (HDD)**
Stores large amounts of data permanently such as operating systems, programs and personal files.

8 **Solid State Drive (SSD)**
Stores data using flash memory. Provides faster boot times and quicker file access than HDD.

9 **Graphics Card (GPU)**
Processes and renders images, video and 3D graphics. Essential for gaming, design and video editing.

10 **Front Panel Connectors**
Connects the case buttons (Power, Reset) and LEDs, as well as USB and audio ports to the motherboard.

11 **CMOS Battery**
Provides power to the BIOS (UEFI) settings and real-time clock when the computer is off or unplugged.

Note: Component types and locations may vary depending on the computer model and case design, but the basic functions remain the same.

Fig 1.1: Internal components of a desktop computer cabinet

ESD PROTECTION

ESD (Electrostatic Discharge) can damage sensitive electronic components. Follow proper ESD protection measures while handling computer hardware.

ESD PROTECTION METHODS

- 1 Wear an anti-static wrist strap. Connect the strap to a grounded point.
- 2 Work on an anti-static mat. The mat safely dissipates static electricity to ground.
- 3 Disconnect power before working inside. Unplug the power cord and all connected cables.

PROPER ESD WORKSTATION SETUP

ESD PROTECTED AREA

Grounded Earth Point

Anti-static Mat

- 4 Touch a grounded metal surface before handling components. This discharges any static electricity from your body.
- 5 Keep components in anti-static bags until needed. Store and transport components in anti-static bags.
- 6 Avoid touching gold contacts / pins on components. Hold components by the edges.

SENSITIVE COMPONENTS – SUSCEPTIBLE TO ESD DAMAGE

CPU Motherboard RAM (Memory) Graphics Card Storage (SSD / HDD) Network / Expansion Cards

WHAT CAN HAPPEN WITHOUT ESD PROTECTION

- Immediate component failure
- Intermittent issues / system instability
- Reduced lifespan of components
- Data loss

ESD DAMAGE

Always follow ESD protection practices to keep your components safe and your system reliable.

IMPLEMENTING EFFECTIVE ESD PROTECTION MEASURES

ESD-safe work area

ESD wrist strap

ESD footwear

Anti-static mat

Reference Table

Component	Function
Motherboard	Main circuit board connecting all components
CPU (Processor)	Executes instructions; performs all calculations
RAM	Temporary high-speed storage for running programs

Component	Function
SMPS (Power Supply)	Converts AC mains to DC voltages for components
HDD/SSD	Permanent storage of data and OS
Cooling Fan/Heatsink	Dissipates heat from CPU/GPU
Graphics Card (GPU)	Processes and outputs video/display data
Expansion Cards	Add extra functionality (sound, network, etc.)

Procedure

1. Switch off the PC and disconnect the mains power cable.
2. Wear an anti-static wrist strap and ground yourself.
3. Open the cabinet side panel using a screwdriver.
4. Visually locate and identify each component shown in Fig. 1.1.
5. Note the model and physical connections of each part.
6. Record observations in the observation table.
7. Close the cabinet after observation is complete.

Precautions

- Always disconnect power before opening the cabinet.
- Handle components by their edges; avoid touching gold contacts.
- Use ESD protection to avoid damaging sensitive parts.

Result

The internal components of the desktop computer were successfully identified and their functions understood.

PRACTICAL 2

Study & Comparison of CISC and RISC Architecture

Aim: To study the CISC and RISC processor architectures and compare their features.

Requirements

- Reference material/datasheets
- Whiteboard/notebook for comparison chart

Theory

Processor architectures are broadly classified as CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer). CISC processors (e.g., Intel x86) use a large set of complex, variable-length instructions that can perform multi-step operations in a single instruction, reducing the number of instructions per program but requiring more clock cycles per instruction. RISC processors (e.g., ARM, RISC-V) use a small set of simple, fixed-length instructions that generally execute in a single clock cycle, relying more on the compiler and a larger set of registers for efficiency.

Diagram

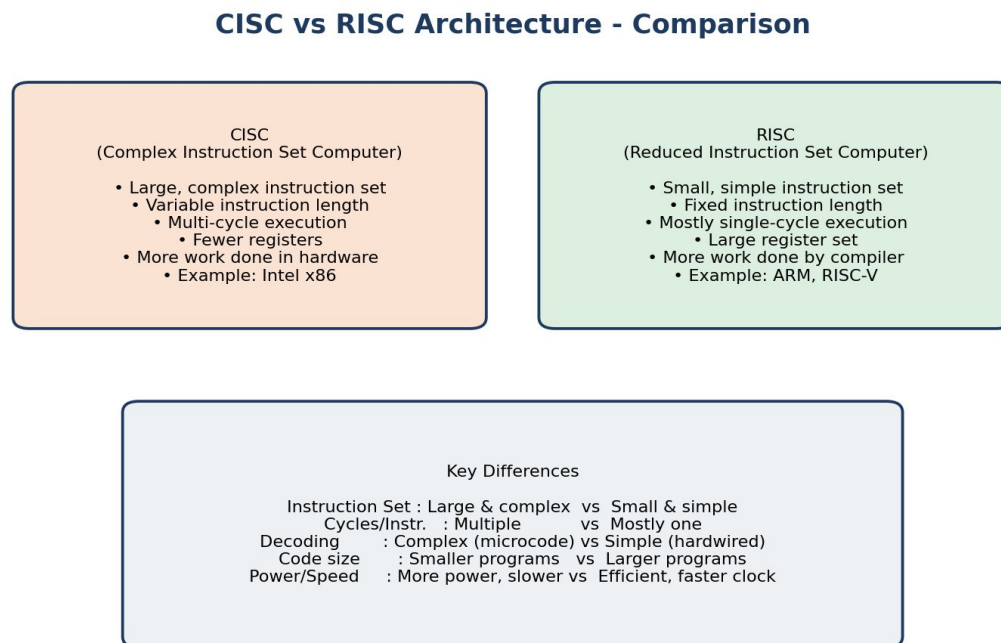


Fig 2.1: CISC vs RISC architecture comparison

Reference Table

Parameter	CISC	RISC
Instruction Set	Large & complex	Small & simple
Instruction Length	Variable	Fixed
Execution Cycles	Multiple cycles/instruction	Mostly 1 cycle/instruction
Registers	Fewer general-purpose registers	Large register file

Parameter	CISC	RISC
Decoding Hardware	Complex (microcode)	Simple (hardwired)
Examples	Intel x86, AMD	ARM, RISC-V, MIPS

Procedure

1. Study the instruction sets of a sample CISC processor (Intel x86 datasheet) and a RISC processor (ARM datasheet).
2. Note the number of instructions, addressing modes, and register count for each.
3. Prepare a comparison chart as shown in Fig. 2.1 and the table above.
4. Discuss real-world applications where CISC vs RISC is preferred (e.g., RISC in mobile/embedded, CISC in desktop/servers).

Precautions

- Refer to authentic/updated datasheets for accurate data.

Result

The CISC and RISC architectures were studied and their differences were tabulated successfully.

PRACTICAL 3**Identify Motherboard Components (CPU Socket, RAM Slots, Chipset, Expansion Slots and Connectors)**

Aim: To identify various components present on a PC motherboard.

Requirements

- Desktop motherboard (removed or in cabinet)
- Screwdriver set
- Torch

Theory

The motherboard is the main printed circuit board of a computer that houses/connects the CPU, memory, chipset, expansion slots and various connectors. Understanding its layout is essential for assembly, upgrade and troubleshooting.

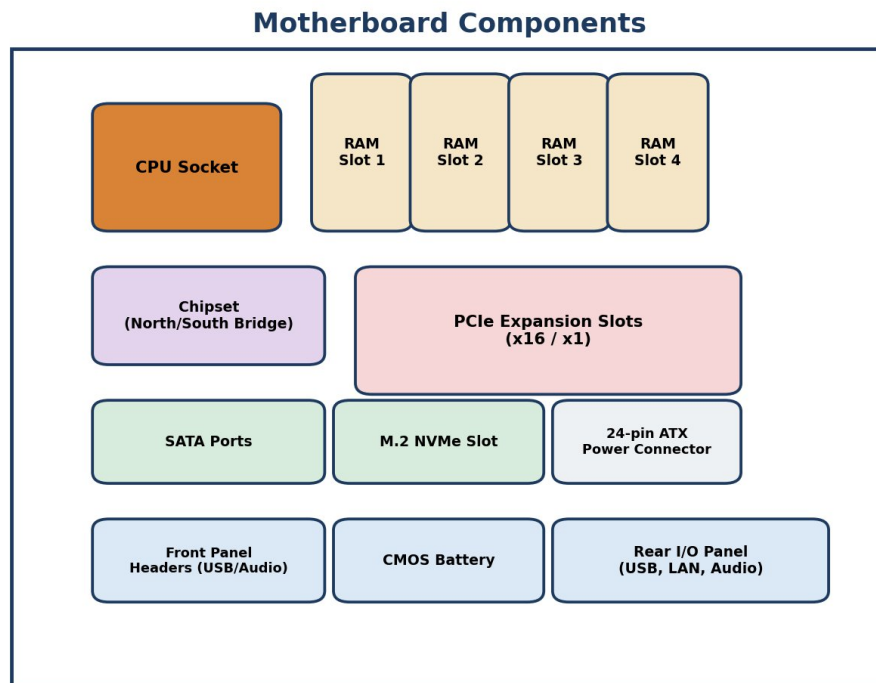
Diagram

Fig 3.1: Layout of a typical PC motherboard

MOTHERBOARD COMPONENTS – IDENTIFY & LOCATE

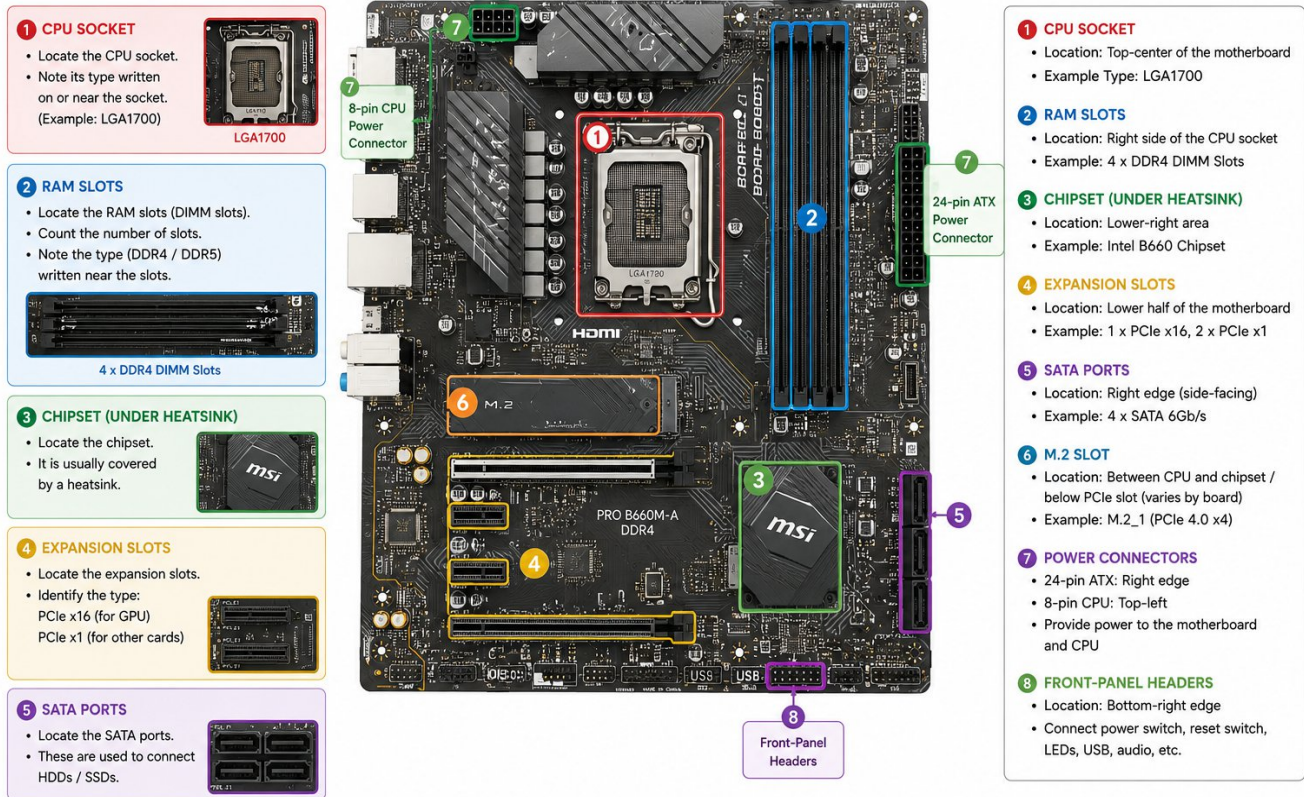


Fig 3.2: Components of a motherboard

Reference Table

Component	Description
CPU Socket	Holds and connects the processor to the motherboard
RAM Slots (DIMM)	Hold memory modules; supports dual/quad channel
Chipset	Manages data flow between CPU, memory and peripherals
PCIe Expansion Slots	For GPU, sound card, network card, etc.
SATA Ports	Connect HDD/SSD/optical drives
M.2 Slot	For NVMe/SATA M.2 SSDs
24-pin ATX Connector	Main power connector from SMPS
Front Panel Header	Connects power/reset switches and LEDs

Procedure

1. Power off and open the cabinet; remove the motherboard if required for closer inspection.
2. Locate the CPU socket and note its type (e.g., LGA, PGA).

3. Identify RAM slots and note their number and type (DDR3/DDR4/DDR5).
4. Locate the chipset (usually under a heatsink) and expansion slots (PCIe x16/x1).
5. Identify SATA ports, M.2 slot, power connectors and front-panel headers.
6. Record the details in the observation table.

Precautions

- Handle the motherboard by its edges.
- Avoid flexing the board to prevent hairline cracks.
- Use ESD protection.

Result

Various components of the motherboard were successfully identified and their functions noted.

PRACTICAL 4

Observe and List Functions of Front Panel and Back Panel Ports of a PC

Aim: To observe the front panel and rear panel ports of a PC and list their functions.

Requirements

- Desktop PC (assembled)
- Notebook for recording observations

Theory

A PC cabinet provides front panel ports for convenient access (USB, audio) and a rear I/O panel for permanent peripheral connections (display, network, audio, legacy ports). Knowing each port's function helps in correct device connection and troubleshooting.

Diagram

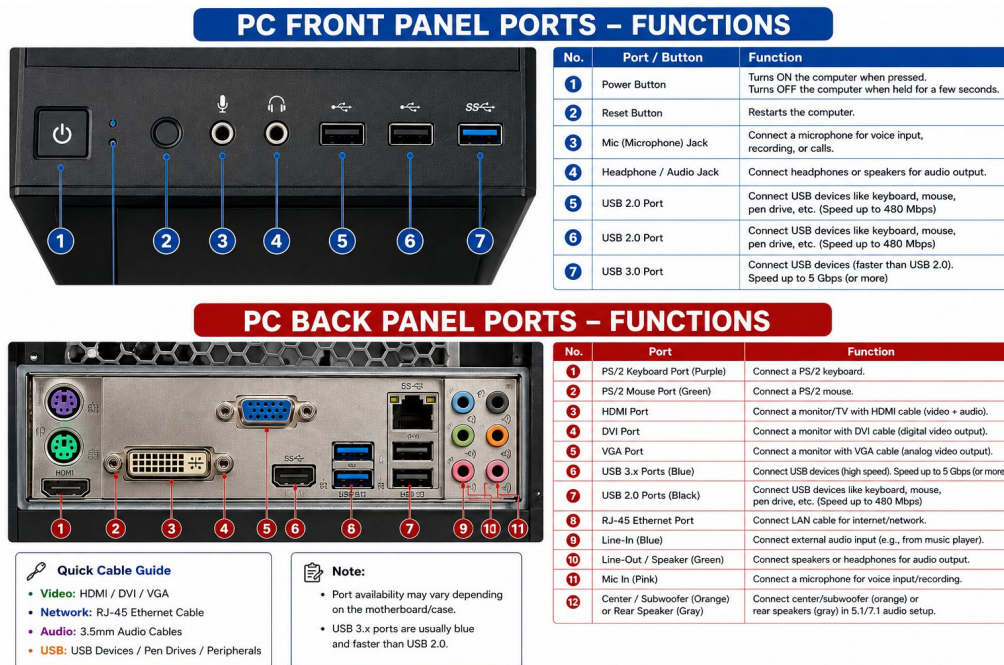


Fig 4.1: Front panel and back panel ports of a PC

Reference Table

Port	Location	Function
USB 2.0/3.0	Front & Back	Connect pen drives, keyboard, mouse, external HDD
Audio Jack (Mic/Headphone)	Front & Back	Connect microphone/speaker/headphone
Power/Reset Button	Front	Power ON/OFF and restart the system
PS/2 Ports	Back	Legacy keyboard/mouse connection
VGA/DVI/HDMI/DisplayPort	Back	Connect monitor for video output

Port	Location	Function
RJ-45 LAN Port	Back	Wired network/internet connection
SMPS Power Socket	Back	Mains power input with ON/OFF switch

Procedure

1. Examine the front panel of the PC cabinet and list all visible ports/buttons.
2. Examine the rear I/O panel and list all ports along with their shape/colour coding.
3. Match each port with the corresponding cable/device it accepts.
4. Record observations as per the table above.

Precautions

- Do not force a connector into the wrong port.
- Power off before connecting/disconnecting ports where required.

Result

Front panel and back panel ports were observed and their functions listed successfully.

PRACTICAL 5

Study SMPS Connectors and Measure Output Voltages Using a Digital Multimeter

Aim: To study different SMPS (Switch Mode Power Supply) connectors and measure their output voltages using a digital multimeter.

Requirements

- SMPS unit
- Digital multimeter
- Paper clip (for PSU tester method)
- Safety gloves

Theory

An SMPS converts AC mains supply into regulated DC voltages (+3.3V, +5V, +12V, -12V, +5VSB) required by different PC components. Each connector type (24-pin ATX, EPS12V, PCIe, SATA, Molex) carries a specific set of voltage lines identified by wire colour.

Diagram

SMPS CONNECTORS

These are the common connectors coming out of SMPS (Switch Mode Power Supply) and their uses.

1. 24-Pin ATX Motherboard Connector	2. 4+4 Pin CPU Power Connector	3. 6+2 Pin PCIe (VGA) Connector	4. SATA Power Connector	5. Peripheral / Molex (4-Pin) Connector	6. Floppy Drive Connector
					
Connects to the motherboard. Provides main power to the system.	Provides power to the CPU on the motherboard.	Provides power to the Graphics Card (GPU).	Powers SATA devices like HDD, SSD, SATA DVD drive etc.	Powers older IDE HDDs, optical drives, case fans, pump etc.	Powers old floppy drives. (rarely used now)

MEASURE OUTPUT VOLTAGES USING A DIGITAL MULTIMETER



How to Measure

1. Turn OFF the power supply and unplug it from AC.
2. Set the multimeter to DC Voltage (V_{DC}).
3. Insert the black probe (COM) into any black wire (GND).
4. Touch the red probe to the desired colored wire of the connector.
5. Read the voltage on the multimeter display.




Black Probe
(to any Black Wire)


Red Probe
(to desired wire)

Common SMPS Output Voltages		
Wire Color	Voltage	Use
Yellow	+12V	CPU, GPU, Motors, Fans
Red	+5V	Motherboard, Logic circuits
Orange	+3.3V	Motherboard, RAM, Chipset
Blue	-12V	Serial ports, legacy circuits
Purple	+5VSB	Standby power, Wake-on-LAN
Black	0V (GND)	Ground / Negative
Green	PS_ON#	Power supply ON/OFF control
Gray	PWR_OK	Power OK signal

Note: Voltage values may vary slightly (±5%). Always measure with care and avoid short circuits.

⚠ CAUTION

- Do not open the SMPS cover.
- High voltages are present inside.
- Use proper safety precautions.

Typical Tolerances (±)

- +12V → 11.40V to 12.60V
- +5V → 4.75V to 5.25V
- +3.3V → 3.14V to 3.47V
- -12V → -10.80V to -13.20V
- +5VSB → 4.75V to 5.25V

Example Readings

- Yellow (to Black) = 12.18 V
- Red (to Black) = 5.06 V
- Orange (to Black) = 3.32 V
- Blue (to Black) = -12.15 V
- Purple (to Black) = 5.10 V

Important Tips

- Always connect black probe to GND (black wire).
- Use sharp probe tips + steady hands for accurate reading.
- If voltage is out of range, the SMPS or connected device may have a problem.

Fig 5.1: SMPS connectors and multimeter measurement points

SMPS Connector Voltage Reference

Wire Colour	Voltage Line	Expected Value
Yellow	+12V	11.4V to 12.6V
Red	+5V	4.75V to 5.25V
Orange	+3.3V	3.14V to 3.47V
Black	Ground (COM)	0V (reference)
Purple	+5VSB (standby)	4.75V to 5.25V
Green	PS_ON	0V when ON, 5V when OFF

Procedure

1. Set the digital multimeter to DC voltage mode (20V range).
2. Short the green PS_ON wire to a black ground wire (paper-clip test) or switch the PC ON to activate the SMPS.
3. Place the black multimeter probe on any black (ground) wire.
4. Place the red probe on the yellow, red and orange wires one at a time, and note the readings.
5. Repeat measurement on SATA and Molex connectors similarly.
6. Compare the measured values with the expected values in the table and note pass/fail.

Precautions

- Never open the SMPS casing — it stores dangerous voltage even when unplugged.
- Ensure correct multimeter mode (DC, not AC) before measuring.
- Do not short unrelated wires together.

Result

SMPS connectors were studied and output voltages were measured and verified within tolerance using a digital multimeter.

PRACTICAL 6**Install and Remove RAM Modules and Verify Detection in BIOS/UEFI**

Aim: To install and remove a RAM module in a desktop PC and verify its detection in BIOS/UEFI.

Requirements

- Desktop PC
- RAM module(s)
- Anti-static wrist strap
- Screwdriver

Theory

RAM (Random Access Memory) is volatile memory used to temporarily hold data and instructions for the CPU. Correct installation in the DIMM slot and BIOS-level verification confirms that memory is recognised and functioning.

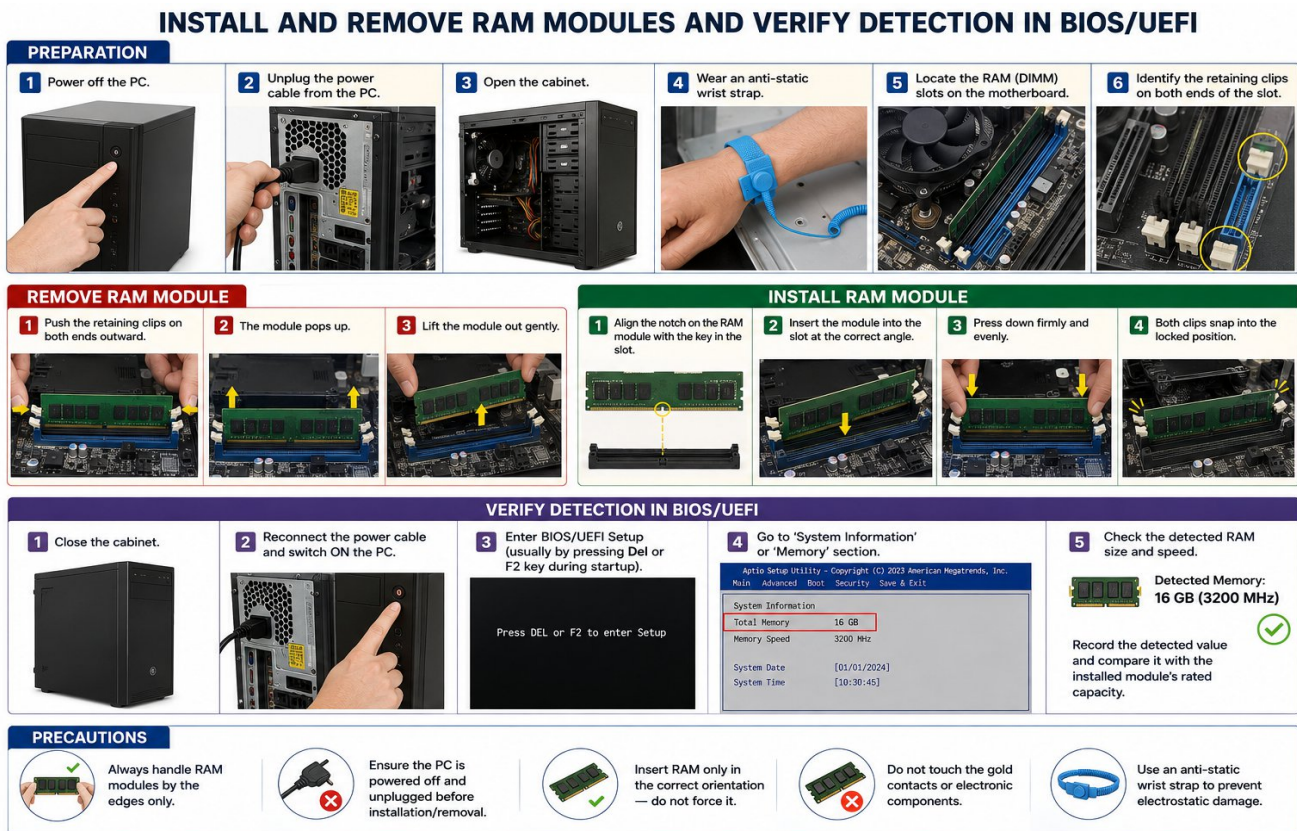
Diagram

Fig 6.1: Steps for installing/removing a RAM module

Procedure

1. Power off the PC, unplug the power cable and open the cabinet.
2. Wear an anti-static wrist strap.
3. To remove: push the retaining clips on both ends of the DIMM slot outward until the module pops up, then lift it out.
4. To install: align the notch on the RAM module with the key in the slot and insert at the correct angle.

5. Press down firmly and evenly until both clips snap into the locked position.
6. Close the cabinet, reconnect power, and switch ON the PC.
7. Enter BIOS/UEFI setup (usually Del/F2 key at startup) and check the 'System Information' or 'Memory' section for detected RAM size and speed.
8. Record the detected value and compare it with the installed module's rated capacity.

Precautions

- Always handle RAM modules by the edges only.
- Ensure the PC is powered off and unplugged before installation/removal.
- Insert RAM only in the correct orientation — do not force it.

Result

The RAM module was installed/removed successfully and correctly detected in BIOS/UEFI.

PRACTICAL 7

Install HDD/SSD and Connect SATA/NVMe Interface Cables

Aim: To install a HDD/SSD in a PC and connect it using SATA or NVMe interfaces.

Requirements

- Desktop PC with free drive bay/M.2 slot
- SATA HDD/SSD or NVMe SSD
- SATA data cable
- Screwdriver

Theory

Storage drives can be connected via the SATA interface (using separate data and power cables) or the newer M.2/NVMe interface (which plugs directly into the motherboard and uses the PCIe bus for much higher speed).

Diagram

INSTALL HDD/SSD AND CONNECT SATA/NVMe INTERFACE CABLES

PROCEDURE

1 Power off the PC and open the cabinet.



2 For SATA: Mount the drive in an available drive bay and secure with screws.



3 Connect one end of the SATA data cable to the drive and the other end to a SATA port on the motherboard.

To Drive



To Motherboard



4 Connect the SATA power connector from the SMPS to the drive.



FOR NVMe (M.2 SSD)

1 Locate the M.2 slot on the motherboard.



2 Insert the SSD at a 30° angle.



3 Press down flat and secure with the retaining screw.



5 Close the cabinet.



6 Power ON the system.

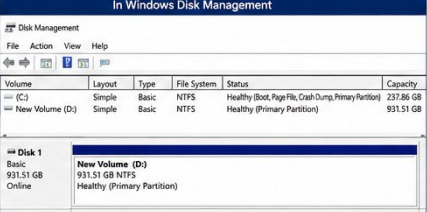


7 Enter BIOS/UEFI or Disk Management (in OS) to verify the drive is detected.

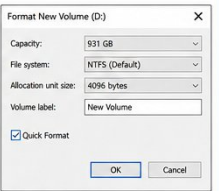
In BIOS/UEFI



In Windows Disk Management



8 Partition and format the new drive if required before use.



PRECAUTIONS

⚠ Handle drives carefully — avoid shocks/drops, especially HDDs.



⚠ Ensure the SATA/power cables are firmly seated.

Correct



Incorrect



⚠ Insert the M.2 SSD at the correct angle to avoid pin damage.

Correct Angle



Wrong Angle (May damage pins)



QUICK REFERENCE

- SATA Data Cable → Motherboard SATA Port
- SATA Power Cable → From SMPS
- M.2 NVMe SSD → M.2 (PCIe/NVMe) Slot
- Always secure drives with screws.
- Verify detection in BIOS/UEFI or OS.

Fig 7.1: SATA and NVMe drive installation

Procedure

1. Power off the PC and open the cabinet.
2. For SATA: Mount the drive in an available drive bay and secure with screws.
3. Connect one end of the SATA data cable to the drive and the other end to a SATA port on the motherboard.
4. Connect the SATA power connector from the SMPS to the drive.

Page 16 of 34

5. For NVMe: locate the M.2 slot on the motherboard, insert the SSD at a 30° angle, press down flat and secure with the retaining screw.
6. Close the cabinet and power ON the system.
7. Enter BIOS/UEFI or Disk Management (in OS) to verify the drive is detected.
8. Partition and format the new drive if required before use.

Precautions

- Handle drives carefully — avoid shocks/drops, especially HDDs.
- Ensure the SATA/power cables are firmly seated.
- Insert the M.2 SSD at the correct angle to avoid pin damage.

Result

The HDD/SSD was successfully installed and detected via the SATA/NVMe interface.

PRACTICAL 8**Configure BIOS/UEFI Settings and Observe Hardware Detection Parameters**

Aim: To access BIOS/UEFI setup, configure basic settings and observe hardware detection parameters.

Requirements

- Desktop PC
- Keyboard

Theory

BIOS (Basic Input Output System) or UEFI (Unified Extensible Firmware Interface) is firmware stored on the motherboard that initializes hardware during boot (POST) and provides a setup utility to configure system parameters such as boot order, date/time, and hardware settings before handing control to the operating system.

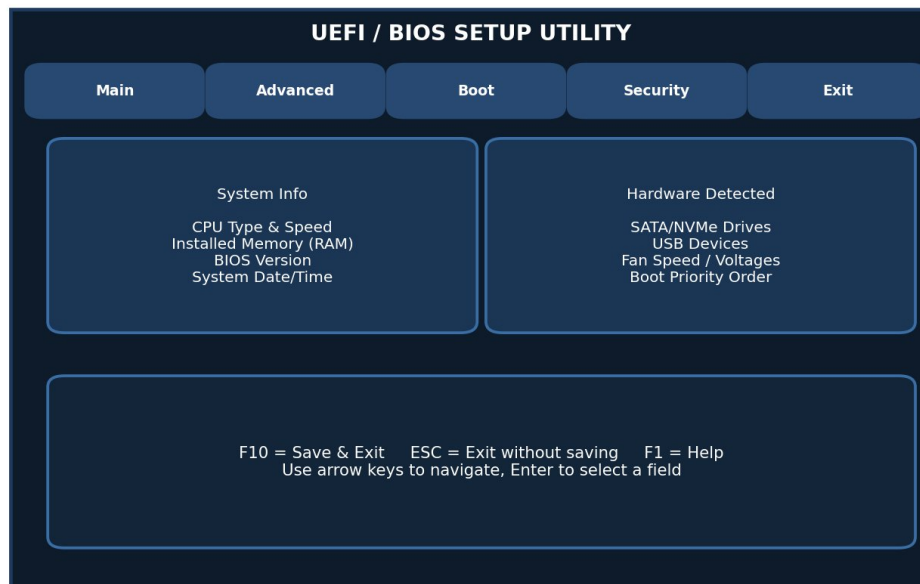
Diagram**BIOS/UEFI Setup - Typical Screen Layout**

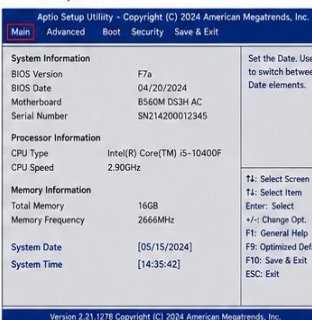
Fig 8.1: Typical BIOS/UEFI setup screen layout

CONFIGURE BIOS/UEFI SETTINGS AND OBSERVE HARDWARE DETECTION PARAMETERS

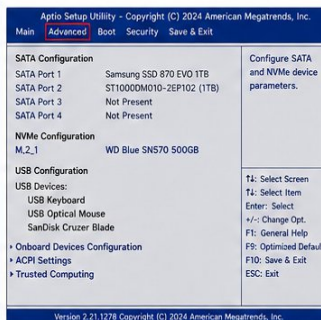
- 1** Restart the PC and press the BIOS/UEFI entry key (Del, F2, F10 or Esc depending on manufacturer) during POST.



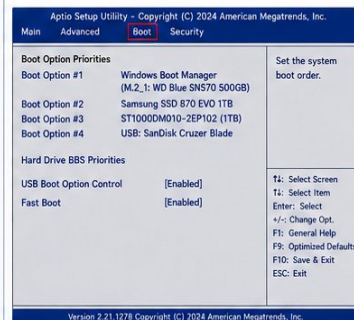
- 2** Navigate to the 'Main' tab and observe system information: CPU type/speed, installed RAM, BIOS version, system date/time.



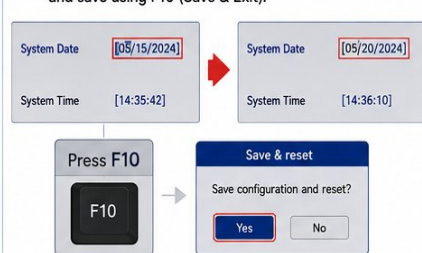
- 3** Navigate to the 'Advanced' tab and observe detected SATA/NVMe drives, USB devices and onboard peripherals.



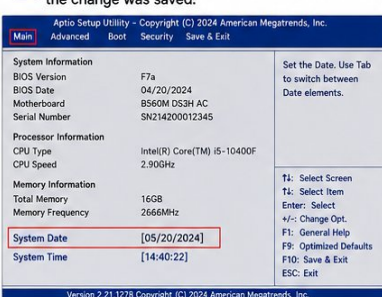
- 4** Navigate to the 'Boot' tab and note/change the boot device priority order.



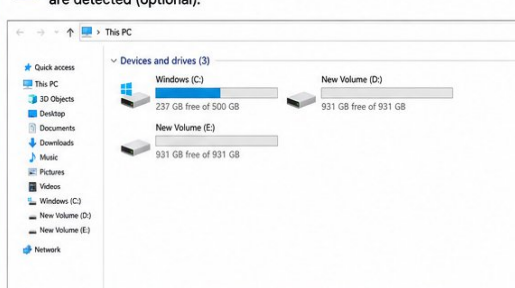
- 5** Make a small safe change (e.g., system date) and save using F10 (Save & Exit).



- 6** Restart and re-enter BIOS to confirm the change was saved.



- 7** Restart the system and boot into the OS to verify the drive(s) are detected (optional).



PRECAUTIONS



Do not change advanced/voltage settings unless certain of the effect. Incorrect settings may cause system instability or boot failure.



Note down original settings before making changes so they can be restored if needed.



Always save changes properly using the Exit menu (usually F10: Save & Exit). Do not power off the system during saving.

QUICK REFERENCE – BIOS ENTRY KEYS

BRAND	BIOS/UEFI ENTRY KEY	Acer	Del or F2
Dell	F2	ASUS	Del or F2
HP	Esc or F10	Gigabyte	Del
Lenovo	F1 or F2	MSI	Del



Post Screen Tip: Watch the POST screen carefully. The BIOS entry key is usually displayed briefly.

Procedure

- Restart the PC and press the BIOS/UEFI entry key (Del, F2, F10 or Esc depending on manufacturer) during POST.
- Navigate to the 'Main' tab and observe system information: CPU type/speed, installed RAM, BIOS version, system date/time.
- Navigate to the 'Advanced' tab and observe detected SATA/NVMe drives, USB devices and onboard peripherals.
- Navigate to the 'Boot' tab and note/change the boot device priority order.
- Make a small safe change (e.g., system date) and save using F10 (Save & Exit).
- Restart and re-enter BIOS to confirm the change was saved.

Precautions

- Do not change advanced/voltage settings unless certain of the effect.
- Note down original settings before making changes so they can be restored.
- Always save changes properly using the Exit menu.

Result

BIOS/UEFI settings were successfully accessed, configured and hardware detection parameters were observed.

PRACTICAL 9**Identify Different Types of Memory Modules and Storage Devices**

Aim: To identify different types of memory modules and storage devices used in computers.

Requirements

- Sample RAM modules (DDR3/DDR4/DDR5)
- HDD
- SSD (SATA & NVMe)
- USB flash drive
- Optical disc

Theory

Computer memory is classified into primary memory (RAM, ROM, Cache — fast, often volatile) and secondary storage (HDD, SSD, optical media, flash drives — non-volatile, larger capacity but slower).

Diagram

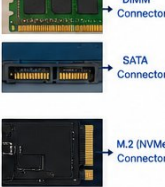
IDENTIFY DIFFERENT TYPES OF MEMORY MODULES AND STORAGE DEVICES

PROCEDURE

1 Examine each sample module/device physically.



2 Note its size, connector type and markings.



DIMM Connector
SATA Connector
M.2 (NVMe) Connector

3 Classify each item as primary or secondary storage.

Primary Storage (Volatile Memory)

- RAM Modules

Secondary Storage (Non-Volatile Memory)

- HDD, SSD, USB Drive
- Memory Card

4 Note approximate capacity and typical use.



5 Record findings in the observation.



TYPES OF MEMORY MODULES

DIMM (DDR4)	DIMM (DDR5)	SO-DIMM (DDR4)	SO-DIMM (DDR5)
 • Size: 133.35 mm (L) • Pins: 288 • Voltage: 1.2V • Use: Desktop PCs	 • Size: 133.35 mm (L) • Pins: 288 • Voltage: 1.1V • Use: Latest Desktop PCs	 • Size: 69.6 mm (L) • Pins: 260 • Voltage: 1.2V • Use: Laptops, Mini PCs	 • Size: 69.6 mm (L) • Pins: 262 • Voltage: 1.1V • Use: Latest Laptops

DIMM – Dual In-line Memory Module | SO-DIMM – Small Outline DIMM

TYPES OF STORAGE DEVICES

HDD (3.5 inch)	SATA SSD (2.5 inch)	M.2 NVMe SSD	USB Flash Drive	SD Card
 • Interface: SATA • Capacity: 500 GB – 16 TB • Use: Mass storage, Backup	 • Interface: SATA III • Capacity: 120 GB – 4 TB • Use: OS, Software, Fast storage	 • Interface: PCIe NVMe • Capacity: 256 GB – 4 TB • Use: High-speed storage, OS, Gaming	 • Interface: USB 2.0/3.0/3.2 • Capacity: 8 GB – 512 GB • Use: File transfer, Backup	 • Interface: SDXC/SDHC • Capacity: 8 GB – 1 TB • Use: Cameras, Phones, Tablets

OBSERVATION TABLE

S. No.	Item / Device	Type / Form Factor	Connector / Interface	Primary or Secondary Storage	Capacity (Approx.)	Typical Use / Application	Markings / Brand
1.		DIMM 288-pin	DDR4 DIMM	Primary (Volatile)	8 GB	Running OS, Applications, Multitasking	Kingston KVR26N19S8/8 1.2V
2.		SO-DIMM (DDR4 RAM) 260-pin	DDR4 SO-DIMM	Primary (Volatile)	4 GB	Laptops, Small Form Factor PCs	Samsung M471A5143DB1 1.2V
3.		3.5 inch	SATA III	Secondary (Non-Volatile)	1 TB	Data storage, Backup, Media	Seagate ST1000DM010 SATA 6Gb/s
4.		2.5 inch	SATA III	Secondary (Non-Volatile)	512 GB	OS, Software, Fast Storage	Samsung 870 EVO 512GB
5.		M.2 2280	PCIe 3.0 x4 NVMe	Secondary (Non-Volatile)	512 GB	High-speed Storage, Gaming, OS	WD Blue SN570 512GB
6.		USB Stick	USB 3.2 Gen 1	Secondary (Non-Volatile)	64 GB	File transfer, Backup, Portable storage	SanDisk Cruzer Blade 64GB
7.		SD Card	SDXC (UHS-I)	Secondary (Non-Volatile)	32 GB	Photos, Videos, Mobile devices	SanDisk Ultra 32GB

PRECAUTIONS

	Handle all memory modules by their edges only. Do not touch the gold contacts.
	Store static-sensitive components in anti-static bags when not in use.
	Avoid physical shocks, drops and excessive pressure, especially for HDDs.
	Keep devices away from liquids, dust and extreme temperatures.
	Ensure connectors are clean and fit properly—do not force any module or device.

Fig 9.1: Types of memory modules and storage devices

Reference Table

Type	Category	Key Feature
DDR3/DDR4/DDR5 RAM	Primary (volatile)	Speed increases, voltage decreases with each generation
ROM/BIOS chip	Primary (non-volatile)	Stores firmware, retains data without power
Cache (L1/L2/L3)	Primary	Extremely fast, built into/near CPU
HDD	Secondary	Magnetic platters, high capacity, slower
SSD (SATA/NVMe)	Secondary	Flash-based, fast, no moving parts
Optical Disc (CD/DVD)	Secondary	Removable, laser-read media
USB Flash Drive	Secondary (removable)	Portable flash storage

Procedure

1. Examine each sample module/device physically and note its size, connector type and markings.
2. Classify each item as primary or secondary storage.
3. Note approximate capacity and typical use of each device.
4. Record findings in the observation table.

Precautions

- Handle all modules by their edges.
- Store static-sensitive components in anti-static bags when not in use.

Result

Different types of memory modules and storage devices were identified and classified successfully.

PRACTICAL 10**Connect and Test Input Devices Such as Keyboard, Mouse and Scanner**

Aim: To connect input devices (keyboard, mouse, scanner) to a PC and test their functioning.

Requirements

- Desktop PC
- USB/PS2 Keyboard
- USB/PS2 Mouse
- Scanner with USB cable and driver CD/software

Theory

Input devices allow the user to provide data/commands to the computer. Common connectors include USB (universal, plug-and-play) and PS/2 (legacy, dedicated port). Correct driver installation ensures the OS recognises and utilises the device fully.

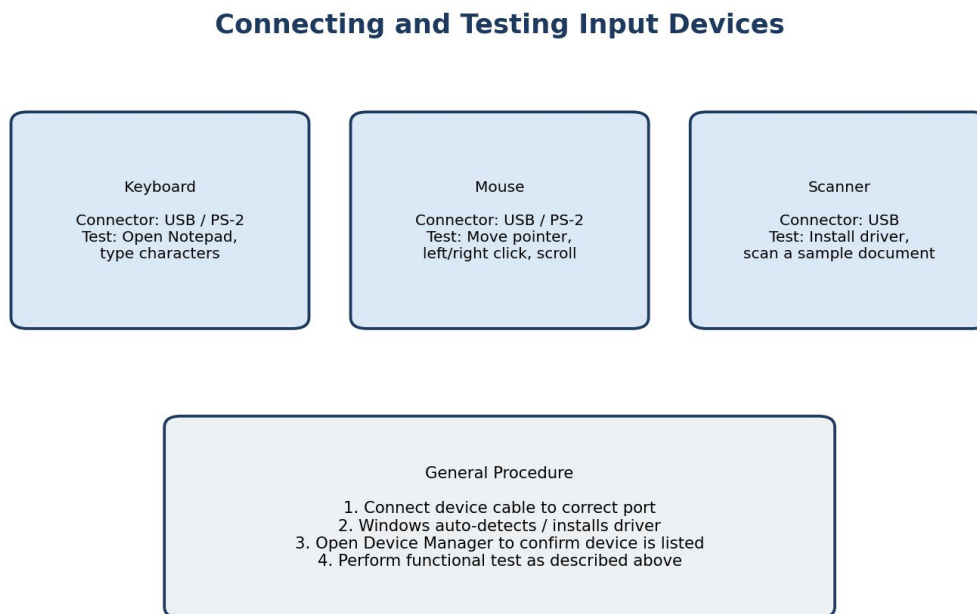
Diagram

Fig 10.1: Connecting and testing input devices

Procedure

1. Connect the keyboard and mouse to available USB ports (or PS/2 ports, matching colour-coding: purple=keyboard, green=mouse).
2. Power ON the PC and allow Windows to auto-detect and install drivers.
3. Test the keyboard by opening a text editor and typing a sample sentence.
4. Test the mouse by moving the pointer, and performing left-click, right-click and scroll actions.
5. Connect the scanner via USB, install its driver/software if not auto-detected.

6. Open the scanner software, place a sample document, and perform a test scan; save the scanned file.
7. Verify all devices are listed correctly under Device Manager.

Precautions

- Ensure the correct port/connector type is used.
- Install manufacturer-provided drivers for scanners for full functionality.
- Do not unplug USB devices abruptly during data transfer.

Result

The keyboard, mouse and scanner were successfully connected and tested for proper functioning.

PRACTICAL 11**Install and Configure a Printer and Perform a Test Print**

Aim: To install a printer on a PC, configure it, and perform a test print.

Requirements

- Desktop PC
- Printer with USB/network cable
- Printer driver software
- A4 paper

Theory

Printer installation involves connecting the hardware, installing the correct driver software matching the printer model and OS version, setting the printer as default, and verifying operation through a test print.

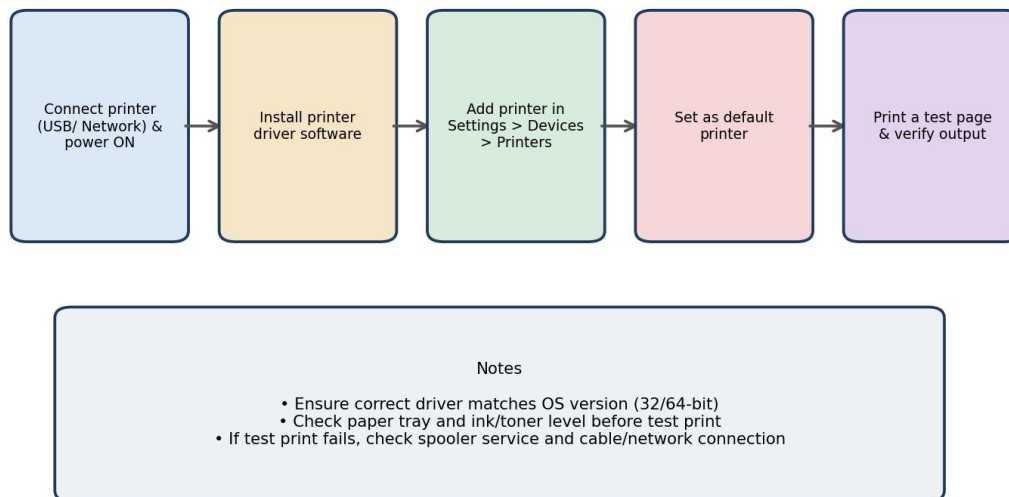
Diagram**Printer Installation and Test Print - Procedure**

Fig 11.1: Printer installation and test print procedure

Procedure

1. Unpack the printer, install ink/toner cartridges and load A4 paper in the tray.
2. Connect the printer to the PC via USB cable or to the network (Wi-Fi/Ethernet).
3. Power ON the printer and PC.
4. Install the printer driver from CD/manufacturer website matching the OS (32-bit/64-bit).
5. Go to Settings > Devices > Printers & Scanners and confirm the new printer is listed; add manually if not auto-detected.
6. Set the installed printer as the default printer.

7. Open a document, select Print, choose the printer, and click Print to perform a test print.
8. Verify the printed output for quality and correct alignment.

Precautions

- Ensure the correct driver version is used for the OS.
- Check ink/toner and paper before printing.
- Do not force paper into the tray; align it properly.

Result

The printer was successfully installed, configured, and a test print was performed.

PRACTICAL 12**Identify Monitor Ports and Configure Display Settings in the Operating System**

Aim: To identify different monitor ports and configure display settings in the operating system.

Requirements

- Desktop PC with monitor
- VGA/DVI/HDMI/DisplayPort cables

Theory

Monitors connect to a PC through video ports such as VGA (analog), DVI (digital/analog), HDMI (digital audio+video) and DisplayPort (high refresh rate digital). Correct display configuration in the OS ensures optimal resolution, orientation and multi-monitor setup.

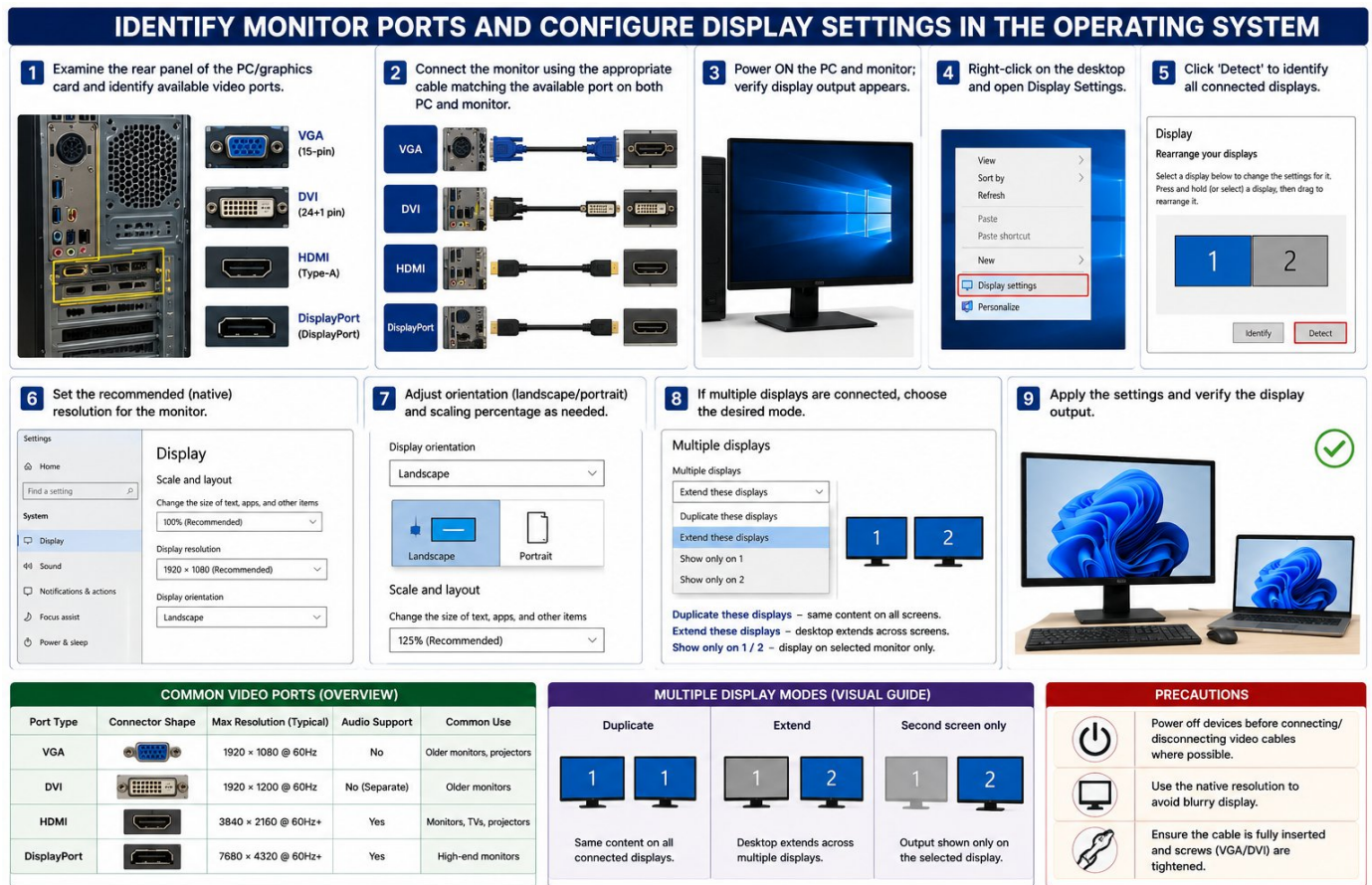
Diagram

Fig 12.1: Monitor ports and OS display settings

Procedure

1. Examine the rear panel of the PC/graphics card and identify available video ports (VGA, DVI, HDMI, DisplayPort).
2. Connect the monitor using the appropriate cable matching the available port on both PC and monitor.
3. Power ON the PC and monitor; verify display output appears.
4. Right-click on the desktop and open Display Settings.

5. Click 'Detect' to identify all connected displays.
6. Set the recommended (native) resolution for the monitor.
7. Adjust orientation (landscape/portrait) and scaling percentage as needed.
8. If multiple displays are connected, choose the desired mode: Duplicate, Extend or Second screen only.

Precautions

- Power off devices before connecting/disconnecting video cables where possible.
- Use the native resolution to avoid blurry display.
- Ensure the cable is fully inserted and screws (VGA/DVI) are tightened.

Result

Monitor ports were identified and display settings were successfully configured in the operating system.

PRACTICAL 13**Perform POST and Interpret Beep Codes and Startup Error Messages**

Aim: To observe the Power-On Self-Test (POST) process and interpret beep codes/startup error messages.

Requirements

- Desktop PC
- Reference chart of BIOS beep codes for the installed BIOS brand (AMI/Award/Phoenix)

Theory

POST (Power-On Self-Test) is a diagnostic sequence run by the BIOS/UEFI immediately after power-on to check core hardware (CPU, RAM, video, keyboard) before the operating system loads. If a fault is detected, the BIOS communicates it through a pattern of beep codes (via the internal speaker) or an on-screen error message, since the video card itself may not yet be initialised.

Diagram

PERFORM POST AND INTERPRET BEEP CODES AND STARTUP ERROR MESSAGES

PROCEDURE

1 Power ON the PC and closely observe the screen and listen for any beep sounds during startup.



Beep... Beep...

2 Note down the number and pattern of beeps (if any) or any error message displayed.



Observation:
1 long beep
3 short beeps
Message: CPU fan error!

3 Compare the beep pattern with the BIOS manufacturer's beep code chart (e.g., AMI/Award/Phoenix).

AMI BIOS BEEP CODES	
Beep Pattern	Meaning
1 short	POST OK
2 short	Parity Check Error
1 long 3 short	No VGA / Video Adapter
Continuous	Memory Error
1 long 2 short	Monitor/Display Error

4 Interpret the meaning of the code/message using the reference table.

INTERPRETATION

Beep Pattern: 1 long 3 short
Meaning: No VGA / Video Adapter
Issue: The system cannot detect the graphics adapter.

Likely Faulty Component:
Graphics card not seated properly, loose cable, or faulty GPU.

5 If an error is found, note the likely faulty component for further troubleshooting.



Likely Faulty Component:
• Graphics card
• Power cable
• Monitor cable / Monitor

6 Record all observations along with the corresponding interpretation.

Record Sheet

Date: 15-05-2024

BIOS: AMI

Beep Pattern: 1 long 3 short

Error Message: CPU fan error!

Interpretation: No VGA / Video Adapter

Likely Faulty Component: Graphics card

Action: Re-seat GPU and check power cable.

BIOS BEEP CODE CHART (EXAMPLES)

AMI BIOS BEEP CODES		AWARD BIOS BEEP CODES		PHOENIX BIOS BEEP CODES	
Beep Pattern	Meaning	Beep Pattern	Meaning	Beep Pattern	Meaning
1 short	POST OK	1 short	POST OK	1 shep	POST OK
2 short	Parity Check Error	1 long 1 short	Motherboard Error	1 long 3 short	System Board Error
1 long 3 short	No VGA / Video Adapter	1 long 2 short	Display Adapter Error	1 long 2 short	System Board Error
1 long 8 short	Display Memory Error	1 long 3 short	Keyboard Error	1 long 3 short	System Board Error
Continuous	Memory Error	1 long 9 short	ROM BIOS Checksum Error	4 short	Memory Error
1 long 2 short	Monitor/Display Error	Continuous	Memory Error	2 short	Parity Error
3 long	Keyboard Error	1 long	DRAM Refresh Failure	1 short repeated	Power Supply Error
1 long 9 short	ROM / BIOS Checksum Error	1 short repeated	Power Supply Problem	3 short	Keyboard Error

Note: Beep codes may vary slightly between motherboard models and BIOS versions. Always refer to the specific motherboard manual.

COMMON STARTUP ERROR MESSAGES

Error Message	Meaning	Likely Faulty Component
CMOS Checksum Error - Default values loaded	BIOS settings are lost or corrupted	CMOS Battery / BIOS settings
CPU Fan Error!	CPU fan not detected or not working	CPU Fan / Fan header / Motherboard
No Bootable Device Found	No OS or boot device is detected	HDD/SSD / Boot order / SATA cable
No VGA Signal	Monitor not getting video signal	Graphics card / Monitor cable / Monitor
Memory Overclock Fail! Press F1 to Run SETUP	Memory settings are unstable	RAM / BIOS settings
Disk Boot Failure	Boot device not accessible	HDD/SSD / Boot order / SATA cable

PRECAUTIONS

Do not ignore repeated abnormal beep codes.

Refer to the correct BIOS brand's chart, as codes differ between manufacturers.

Ensure the internal speaker is connected to the motherboard to hear beep codes.

OBSERVATION RECORD SHEET (EXAMPLE)

S. No.	Date	BIOS Brand	Beep Pattern / Error Message	Interpretation / Meaning	Likely Faulty Component	Action Taken	Result
1	15-05-2024	AMI	1 long 3 short	No VGA / Video Adapter	Graphics Card	Re-seated GPU, Checked power cable	Issue Resolved
2	15-05-2024	AMI	CPU fan error!	CPU fan not detected or not working	CPU Fan / Fan header	Reconnected CPU fan cable	Issue Resolved
3	15-05-2024	AWARD	1 long 2 short	Display Adapter Error	Graphics Card / Monitor cable	Checked cable and monitor	Issue Resolved
4	15-05-2024	AMI	No Bootable Device Found	No OS or boot device is detected	HDD / Boot order	Checked HDD & set correct boot order	Issue Resolved

INTERNAL SPEAKER CONNECTION

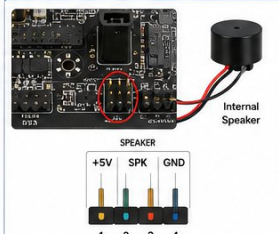


Fig 13.1: POST sequence and beep code interpretation

Reference Table

Beep Pattern (AMI BIOS)	Probable Fault
1 short beep	System OK, normal boot
2 short beeps	RAM parity error

Beep Pattern (AMI BIOS)	Probable Fault
3 short beeps	Base 64K RAM failure
Continuous beep	Power supply/RAM not seated properly
Repeating long beeps	Memory not detected/seated
1 long + 2 short beeps	Video card error

Procedure

1. Power ON the PC and closely observe the screen and listen for any beep sounds during startup.
2. Note down the number and pattern of beeps (if any) or any error message displayed.
3. Compare the beep pattern with the BIOS manufacturer's beep code chart (e.g., AMI/Award/Phoenix).
4. Interpret the meaning of the code/message using the reference table.
5. If an error is found, note the likely faulty component for further troubleshooting.
6. Record all observations along with the corresponding interpretation.

Precautions

- Do not ignore repeated abnormal beep codes.
- Refer to the correct BIOS brand's chart, as codes differ between manufacturers.
- Ensure the internal speaker is connected to the motherboard to hear beep codes.

Result

The POST process was observed and beep codes/error messages were successfully interpreted.

PRACTICAL 14

Troubleshoot Common Hardware Faults Such as No Display, Boot Failure and RAM Errors

Aim: To diagnose and troubleshoot common PC hardware faults: no display, boot failure and RAM errors.

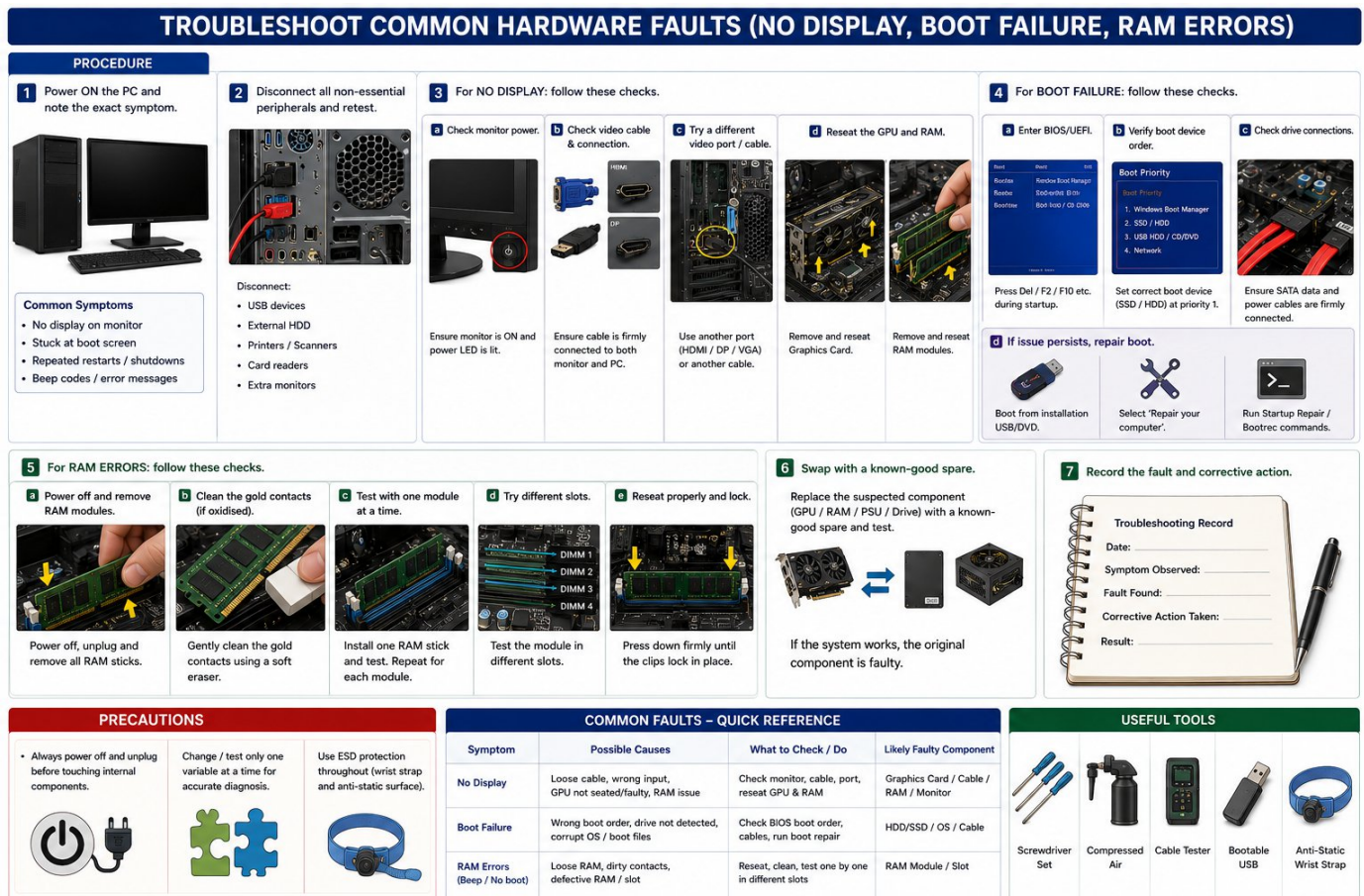
Requirements

- Desktop PC (faulty)
- Screwdriver
- Spare/known-good RAM, cable, GPU (for swap testing)
- Anti-static wrist strap

Theory

Hardware troubleshooting follows a systematic isolation method: observe the exact symptom, disconnect non-essential peripherals, reseal/swap suspected components one at a time and retest, thereby narrowing down the faulty part.

Diagram



PRECAUTIONS

- Always power off and unplug before touching internal components.
- Change / test only one variable at a time for accurate diagnosis.
- Use ESD protection throughout (wrist strap and anti-static surface).

COMMON FAULTS – QUICK REFERENCE

Symptom	Possible Causes	What to Check / Do	Likely Faulty Component
No Display	Loose cable, wrong input, GPU not seated/faulty, RAM issue	Check monitor, cable, port, reseat GPU & RAM	Graphics Card / Cable / RAM / Monitor
Boot Failure	Wrong boot order, drive not detected, corrupt OS / boot files	Check BIOS boot order, cables, run boot repair	HDD/SSD / OS / Cable
RAM Errors (Beep / No boot)	Loose RAM, dirty contacts, defective RAM / slot	Reseat, clean, test one by one in different slots	RAM Module / Slot

USEFUL TOOLS

Screwdriver Set Compressed Air Cable Tester Bootable USB Anti-Static Wrist Strap

Fig 14.1: Troubleshooting flow for common hardware faults

Reference Table

Symptom	Possible Cause	Corrective Action
No Display	Loose/faulty video cable, GPU not seated	Reseat cable/GPU, try a different port/monitor
Boot Failure	Incorrect boot order, corrupted boot sector	Check BIOS boot order, repair/reinstall OS
RAM Errors (beep/BSOD)	RAM not seated, faulty module, dust on contacts	Reseat/clean RAM, test one module at a time

Procedure

1. Power ON the PC and note the exact symptom (no display / stuck at boot / repeated restarts).
2. Disconnect all non-essential peripherals and retest.
3. For no display: check monitor power and cable, try a different video port/cable, reseat the GPU/RAM.
4. For boot failure: enter BIOS and verify boot device order; check HDD/SSD cable connections; run a boot repair from installation media if needed.
5. For RAM errors: power off, reseat each RAM module, clean the gold contacts with an eraser if oxidised, test with a single module at a time in different slots.
6. Swap the suspected component with a known-good spare to confirm the fault.
7. Record the fault found and the corrective action taken.

Precautions

- Always power off and unplug before touching internal components.
- Change/test only one variable at a time for accurate diagnosis.
- Use ESD protection throughout.

Result

Common hardware faults were successfully diagnosed and rectified using systematic troubleshooting.

PRACTICAL 15

Assemble and Disassemble a PC Cabinet Following Safety and ESD Precautions

Aim: To assemble and disassemble a PC cabinet correctly while observing safety and ESD (Electro-Static Discharge) precautions.

Requirements

- PC Cabinet
- Motherboard, CPU, cooler, RAM, PSU, HDD/SSD, GPU
- Screwdriver set
- Anti-static wrist strap/mat

Theory

Assembling a PC requires following a correct sequence to avoid damage and ensure all connections are secure, while ESD precautions protect sensitive electronic components from static-electricity damage during handling.

Diagram



Fig 15.1: PC assembly sequence and safety precautions

Procedure

1. Wear an anti-static wrist strap and work on a grounded/ESD mat; unplug the mains power supply.
2. Install the PSU into the cabinet and secure with screws.

3. Mount the CPU onto the motherboard socket (align notches) and attach the CPU cooler with thermal paste applied.
4. Insert RAM modules into the DIMM slots until clips lock.
5. Fix the motherboard onto the cabinet standoffs and secure with screws.
6. Mount HDD/SSD in drive bays and connect SATA data and power cables.
7. Install the GPU/expansion cards into PCIe slots and secure with screws.
8. Connect front-panel headers (power, reset, LEDs, USB, audio) to the motherboard as per the manual.
9. Connect all power connectors (24-pin ATX, CPU 4/8-pin, PCIe) from PSU to motherboard/GPU.
10. Close the cabinet, connect peripherals, and power ON to verify successful assembly (POST, display, BIOS detection).
11. To disassemble: reverse the above steps — disconnect cables, remove cards/GPU, remove drives, unscrew and remove the motherboard, remove RAM/cooler/CPU, and finally remove the PSU.

Precautions

- Always disconnect the mains power supply before opening the cabinet.
- Wear an ESD wrist strap and avoid working on carpeted surfaces.
- Handle the CPU, RAM and expansion cards by their edges only.
- Apply thermal paste in the correct quantity before mounting the CPU cooler.
- Do not force any connector; verify orientation before insertion.

Result

The PC cabinet was successfully assembled and disassembled following proper safety and ESD precautions.